PHYS 7501:
Particles and Nuclei I

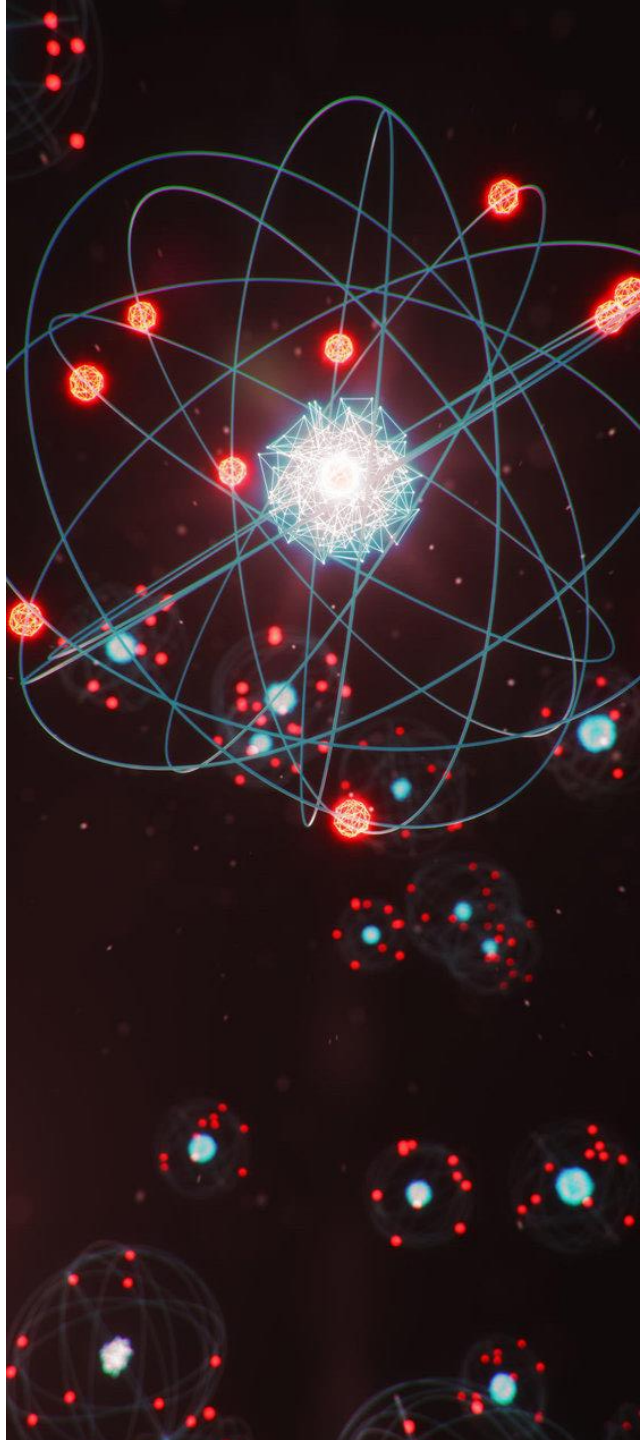
Tues./Thurs. 2 – 3:20PM
Clippinger 105

Dr. Andrea Richard

Edwards Accelerator Lab 204

richarda1@ohio.edu

Available via e-mail & Teams

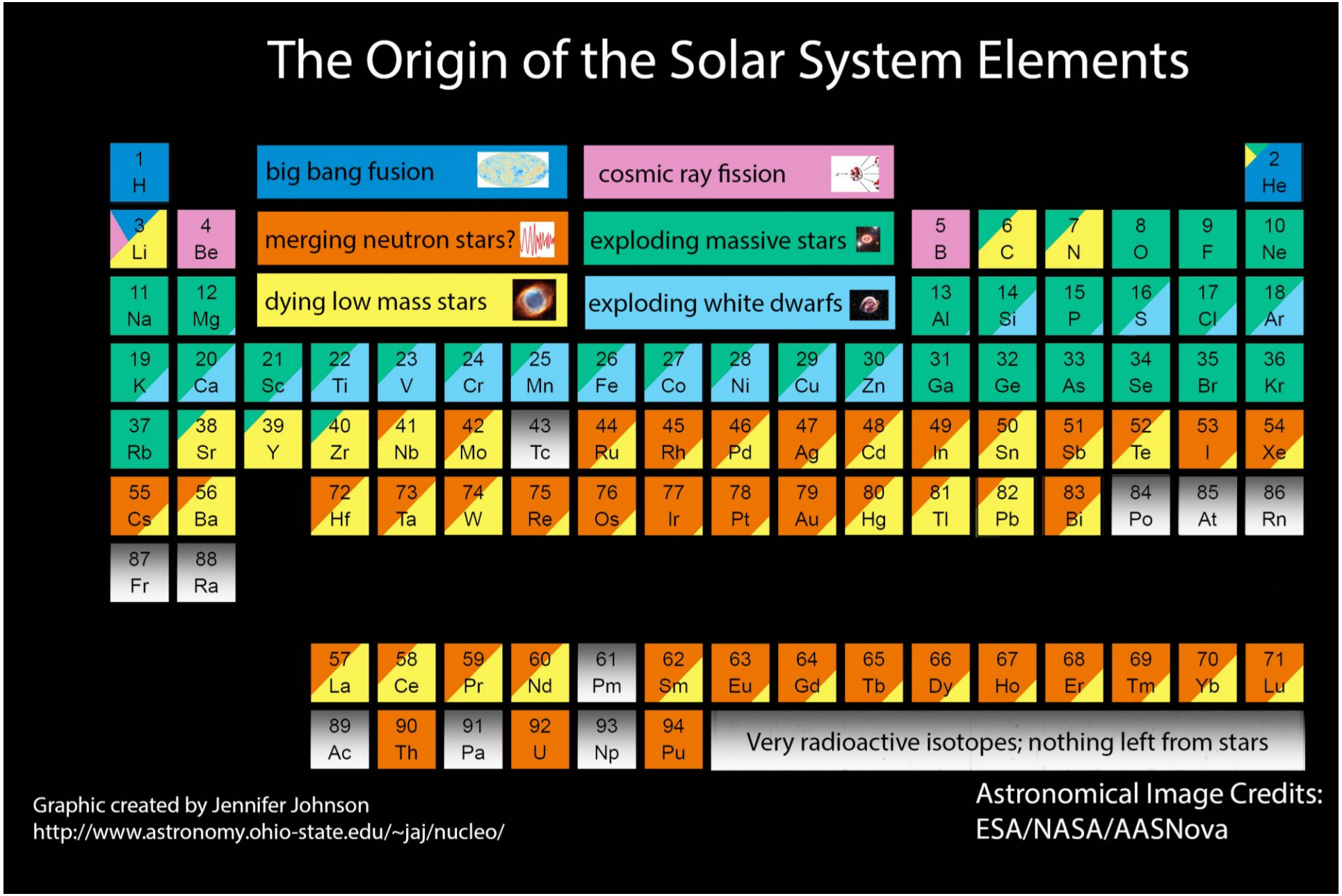


NUCLEAR ASTROPHYSICS

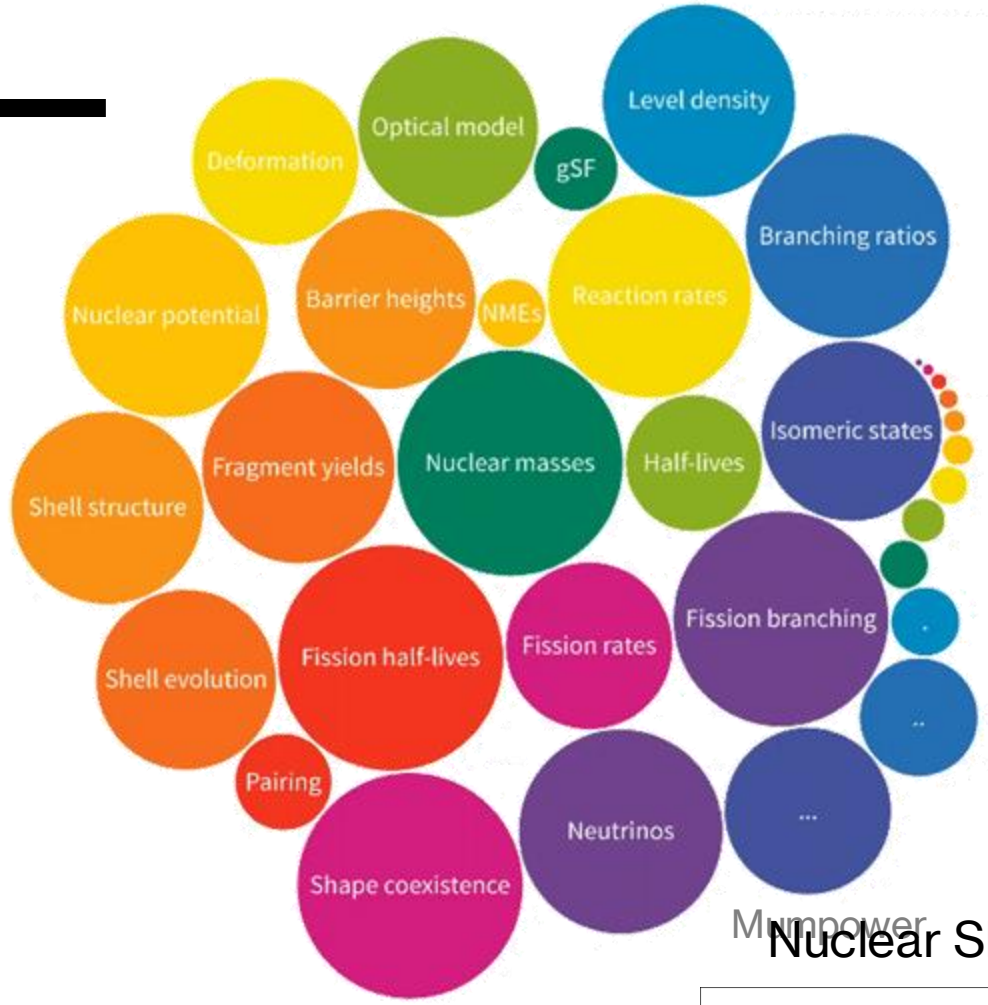
Modern Nuclear Chemistry, Chapter 12

Monday	Tuesday	Wednesday	Thursday	Friday
11/10	11/11	11/12	11/13	11/14
	No Class		**Nuclear Astrophysics** **HW#4 Due**	
11/17	11/18	11/19	11/20	11/21
	Experimental Techniques **Paper Outline Due**		**Nuclear Medicine**	
11/24	11/25	11/26	11/27	11/28
	Nuclear Energy **Draft of Slides Due**		Thanksgiving Break	

Nuclear Astrophysics & the Periodic Table



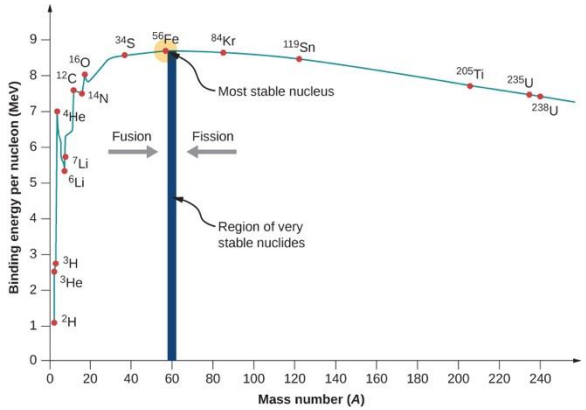
Nuclear Properties in Astrophysics

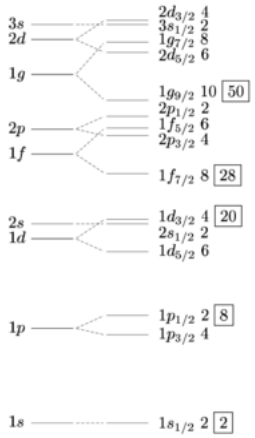




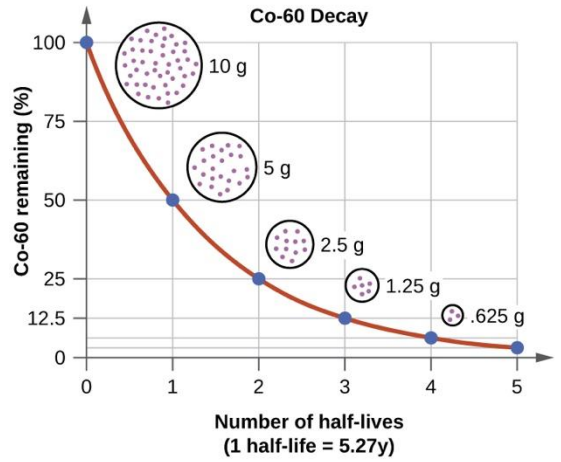
Mass

Binding Energy





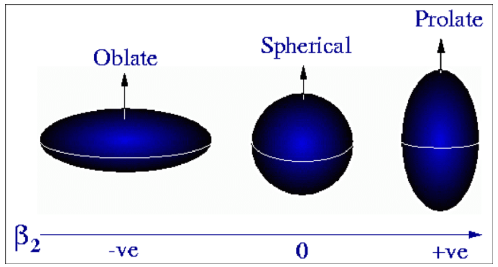
Level structure



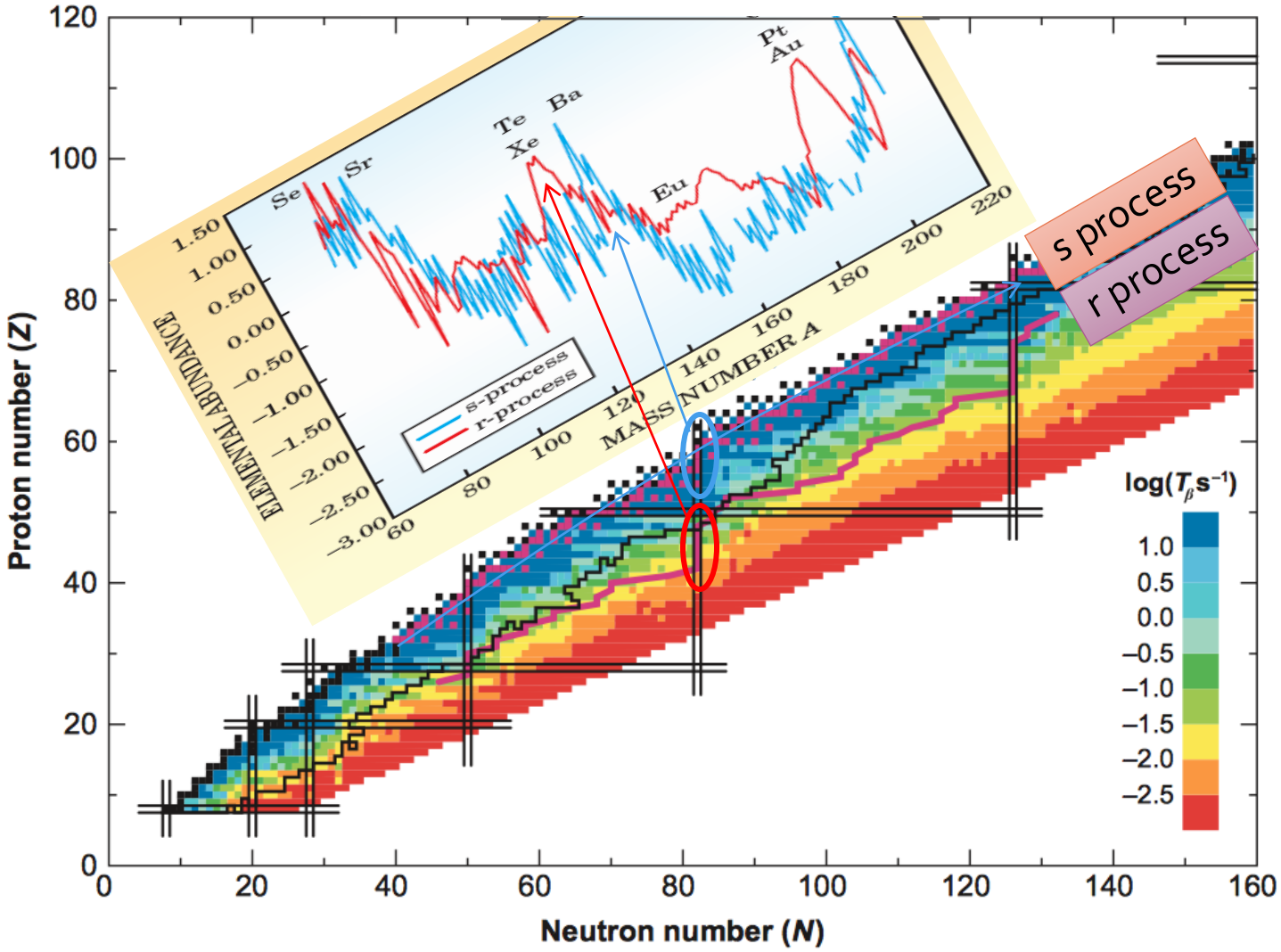
Half-life

Mumpower

Nuclear Shapes



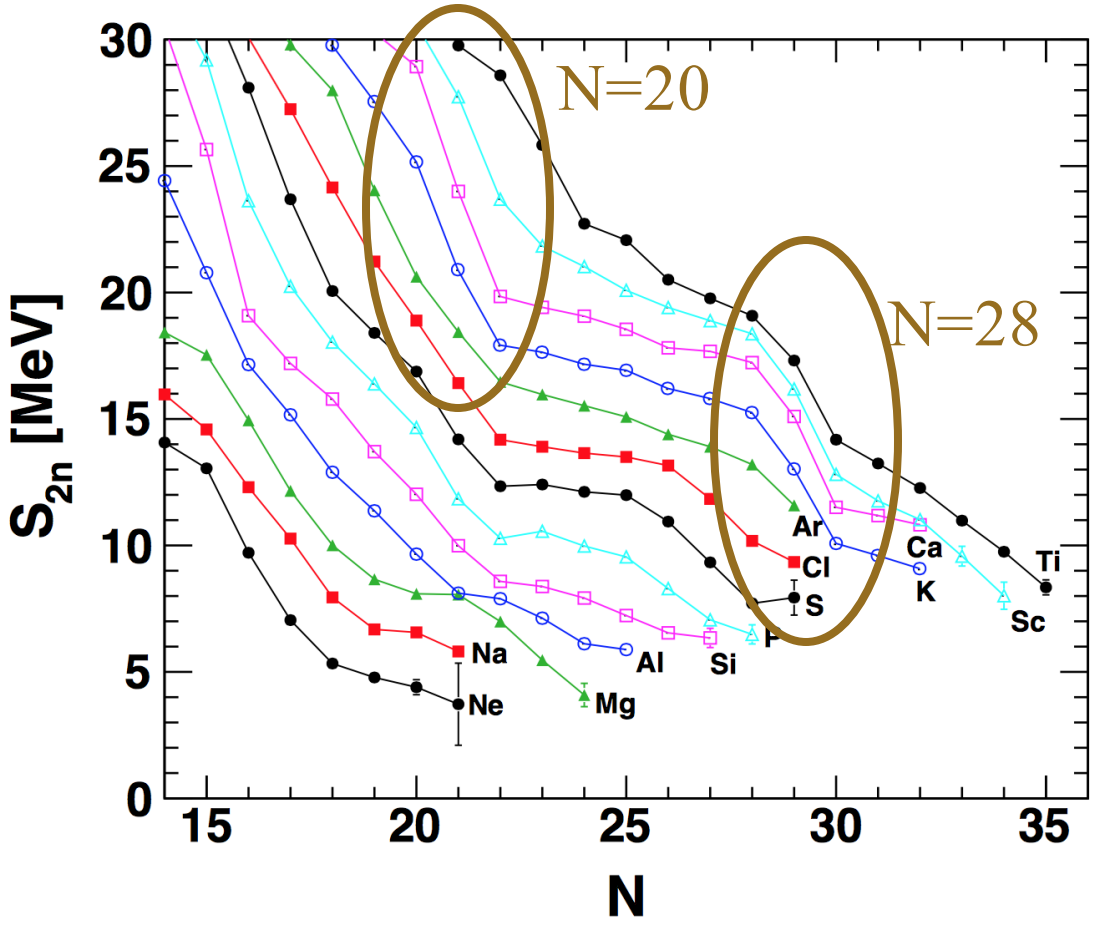
Abundances Strongly Impacted at Magic Numbers

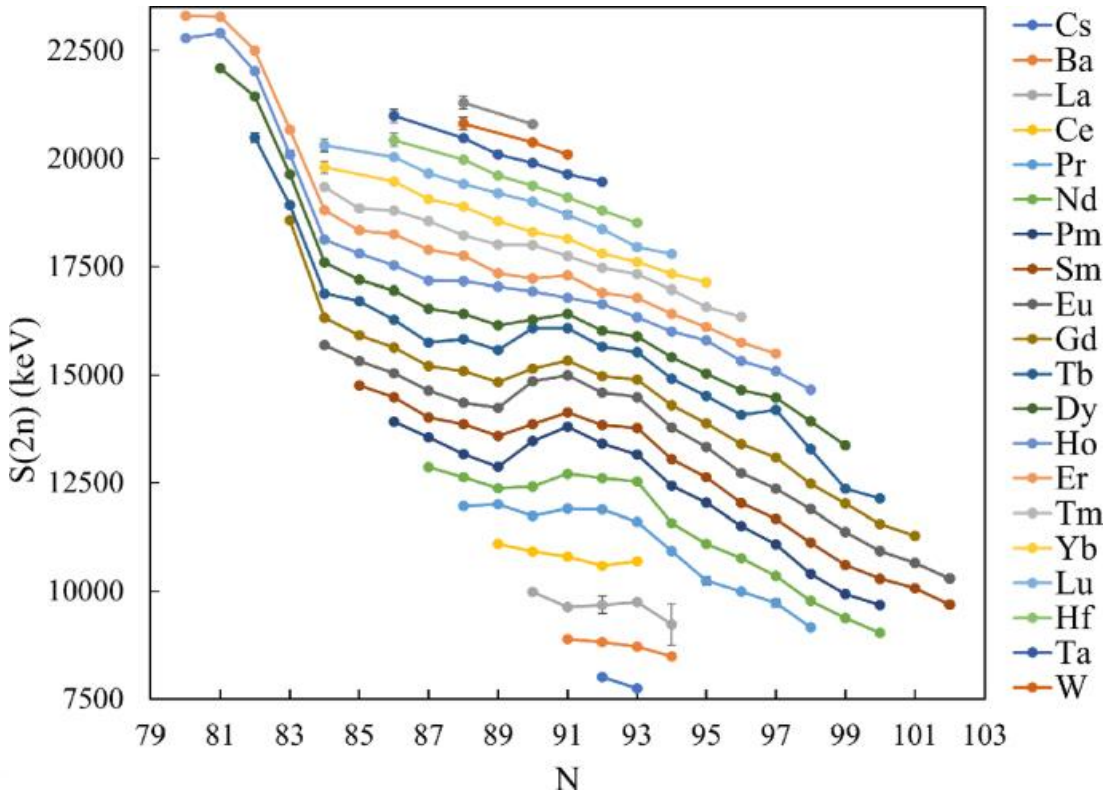


Sneden, C., Cowan, J. J., & Gallino, R., Ann. Rev. Ast. Ap. 46 (2008) 241.

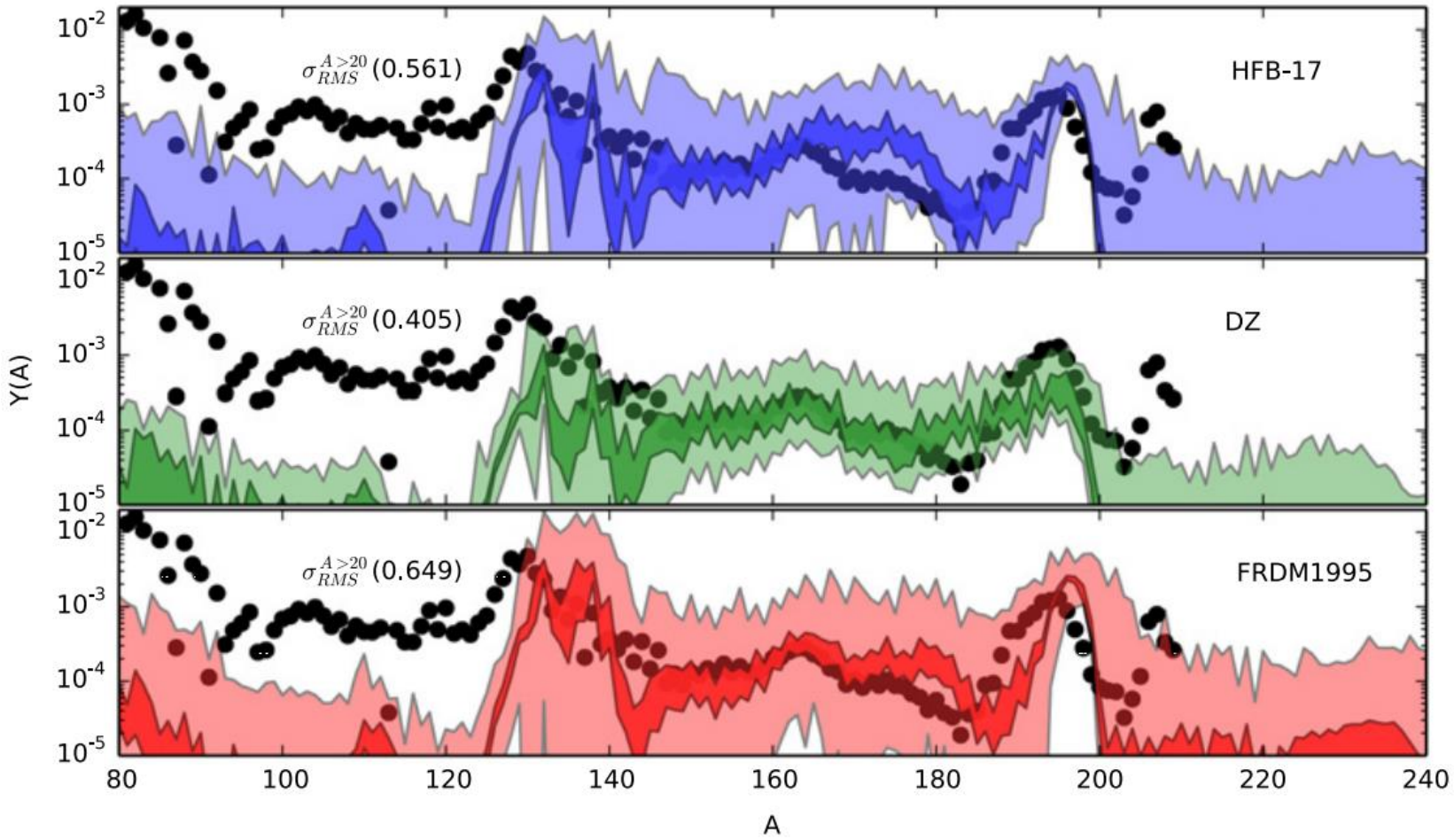
Nuclear Masses

- Strong connection between nuclear mass (binding energy) and shell closures
- Two-neutron separation energies – example





Impact of Nuclear Masses on Astrophysics

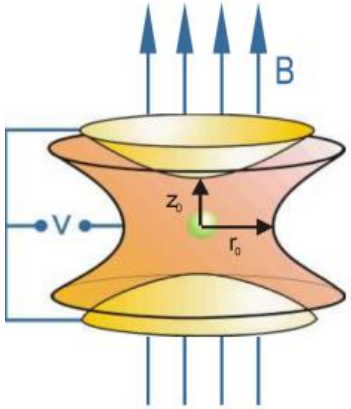


Mumpower *et al.* Prog. Part. Nucl. Phys. **86**, 86 (2016)

Mass Measurements

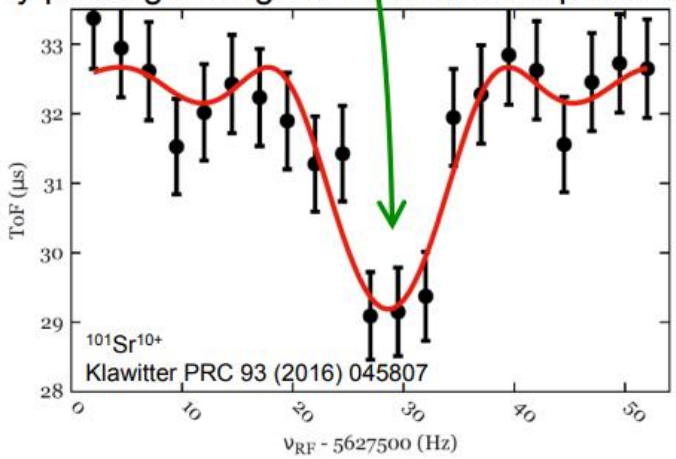
- Masses need to be known to very high precision
- One way to do that is with Penning Traps
- Triple motion in penning trap
- Used for measuring the mass of nuclei with very large accuracy
- Confining particles in electric and magnetic fields

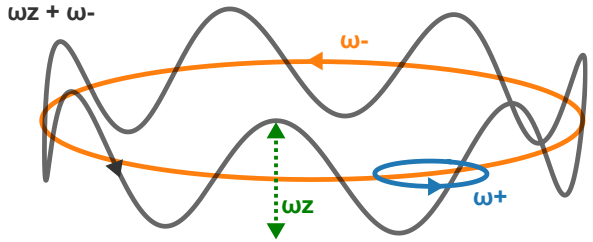




$$\nu_c = \frac{1}{2\pi}\frac{qB}{m}$$

By probing a range of excitation frequencies…

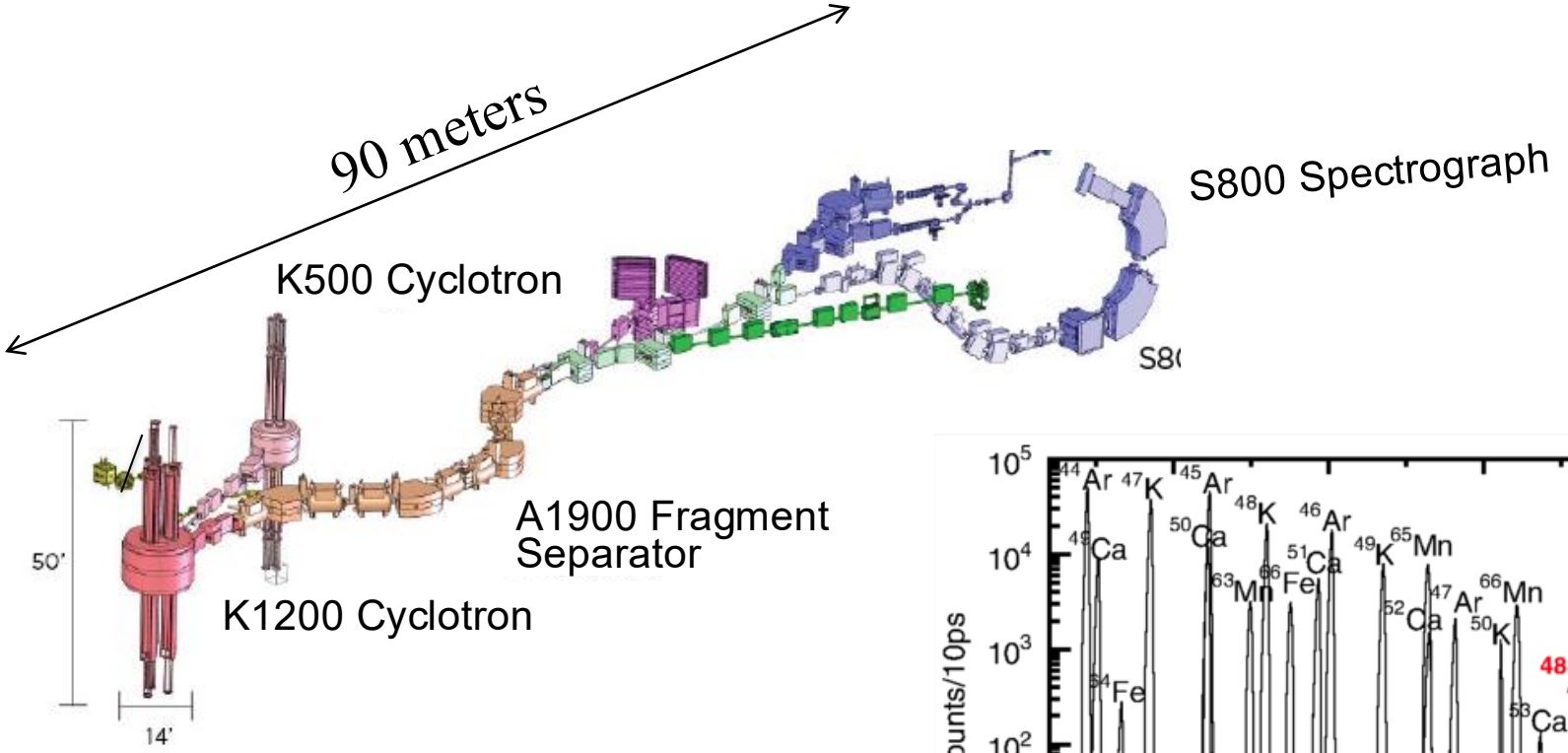


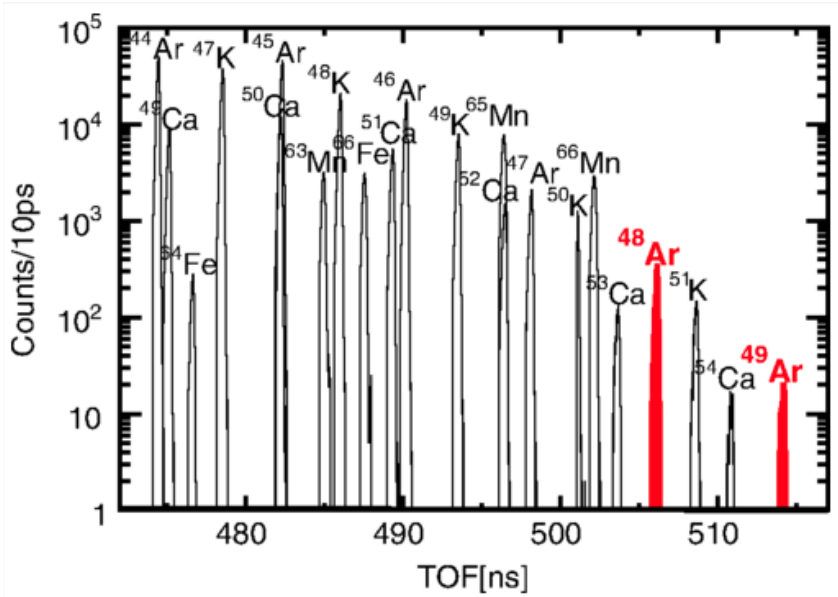


Slide modified from Koote, Orford

Mass Measurements

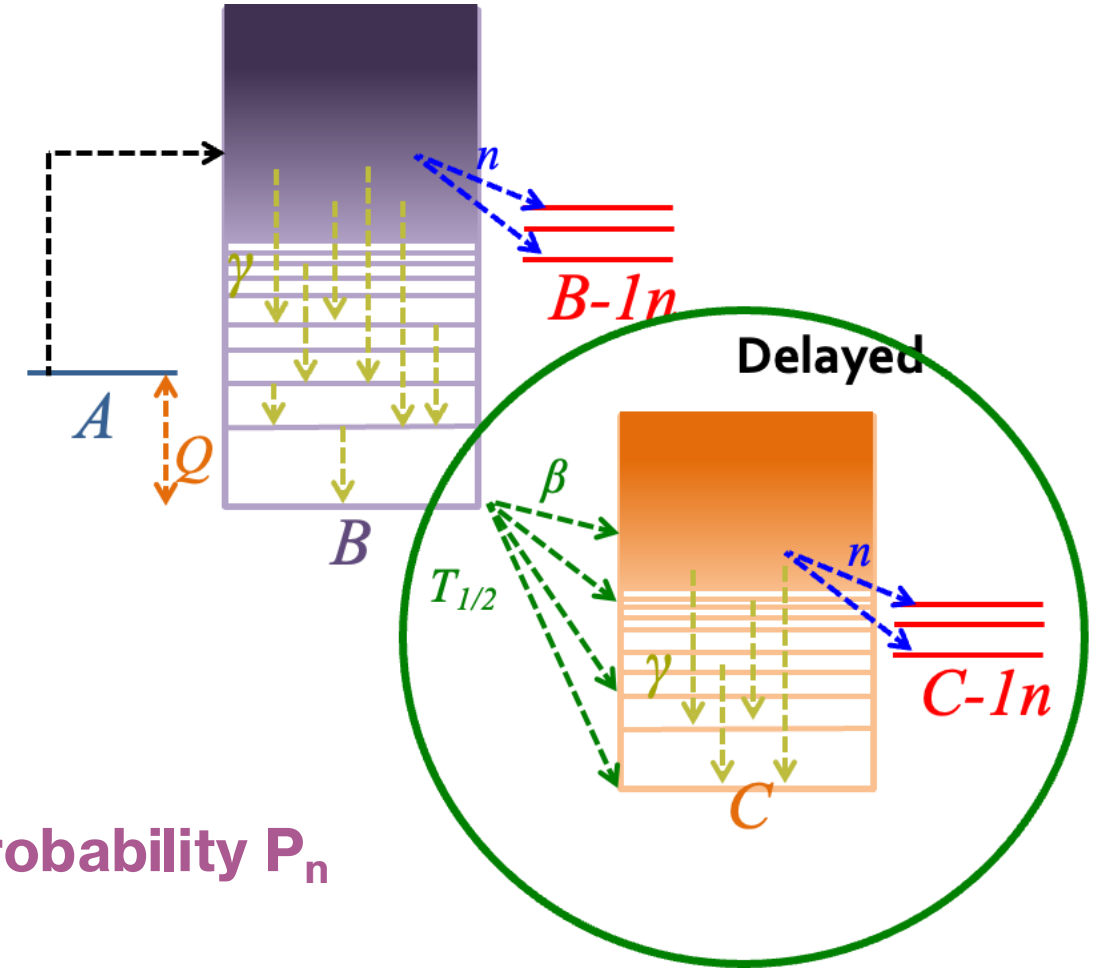
- Time-of-flight can also be used
  - **National Superconducting Cyclotron Laboratory @ MSU**





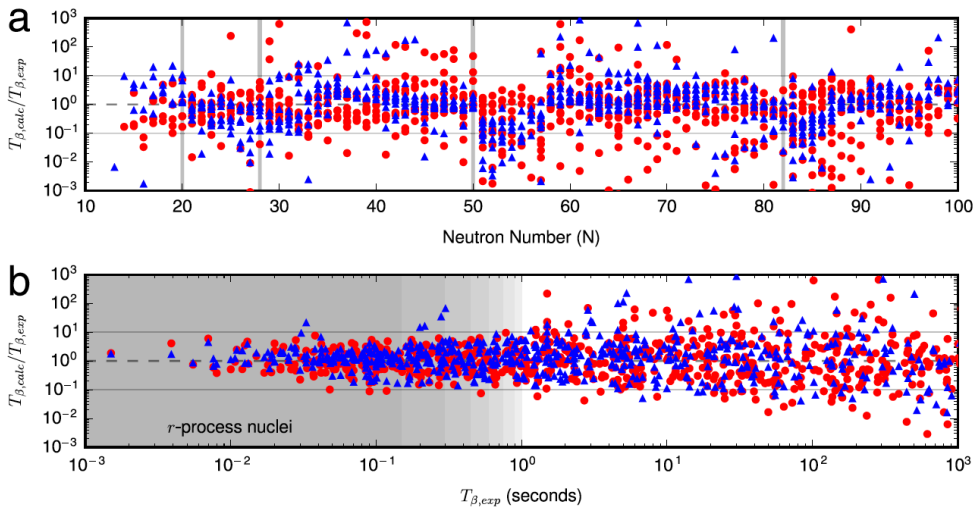
Z. Meisel, S. George, et. al. Prys. Rev. Lett. 114 (2015) 022501

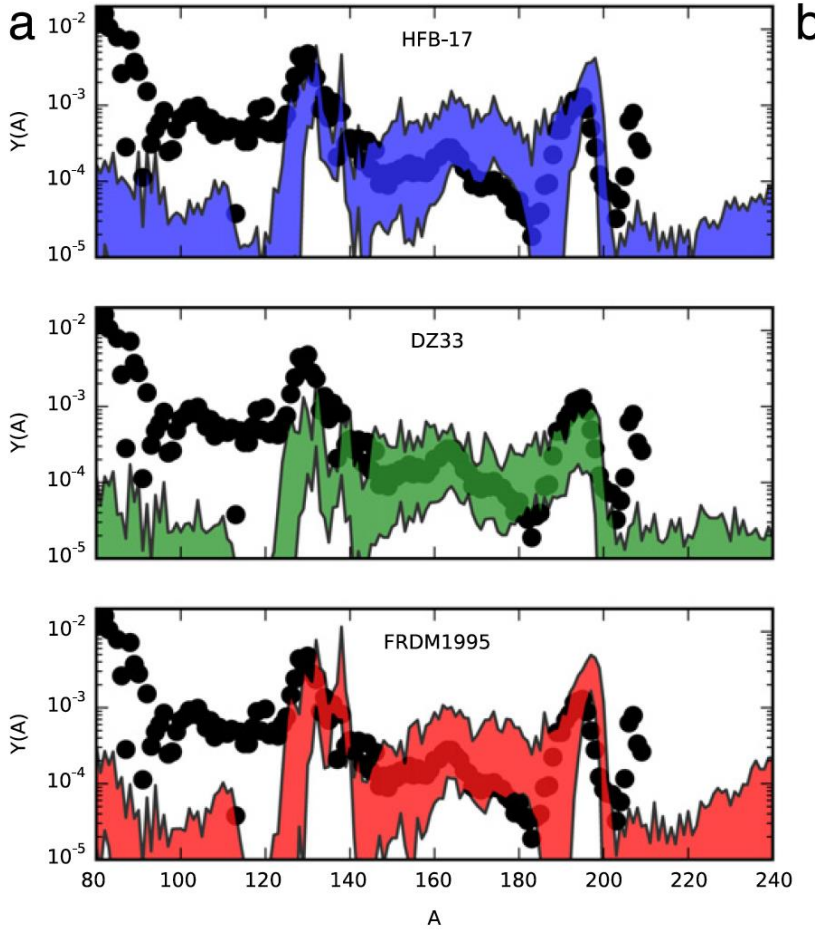
β Decay Properties



- We need to know:
  - Half life $T_{1/2}$
  - β-delayed neutron emission probability P_n
  - β-decay intensity

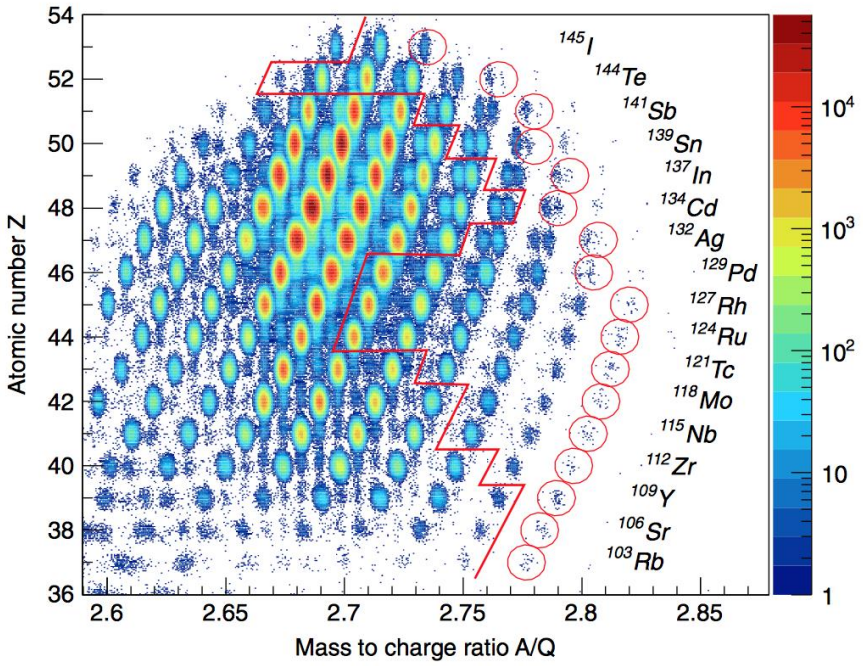
Impact of Half-lives on Astrophysics

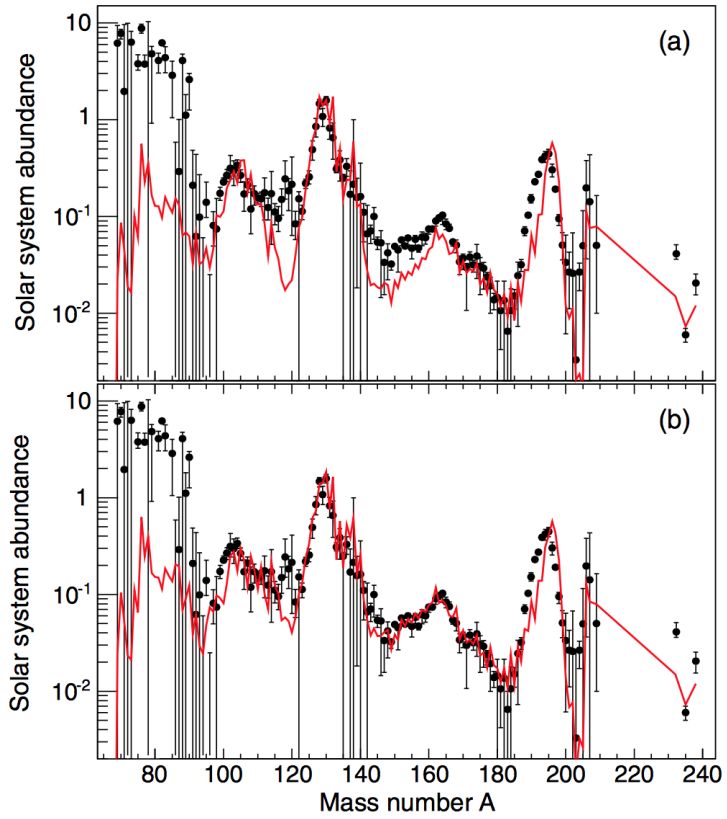




M. Mumpower, et. al. Prog. Nucl. Part. Phys. 86 (2016) 86.

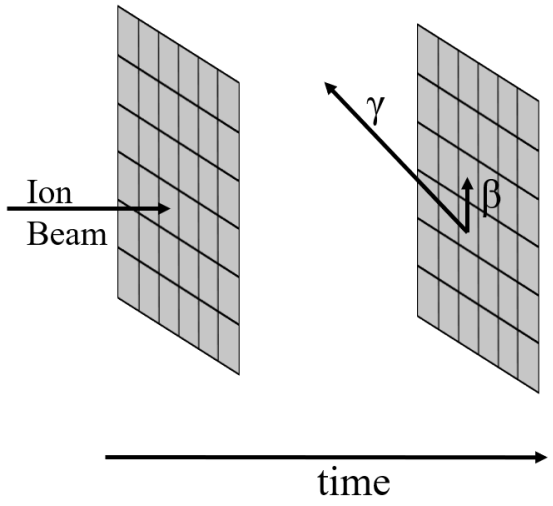
Impact of Half-lives on Astrophysics

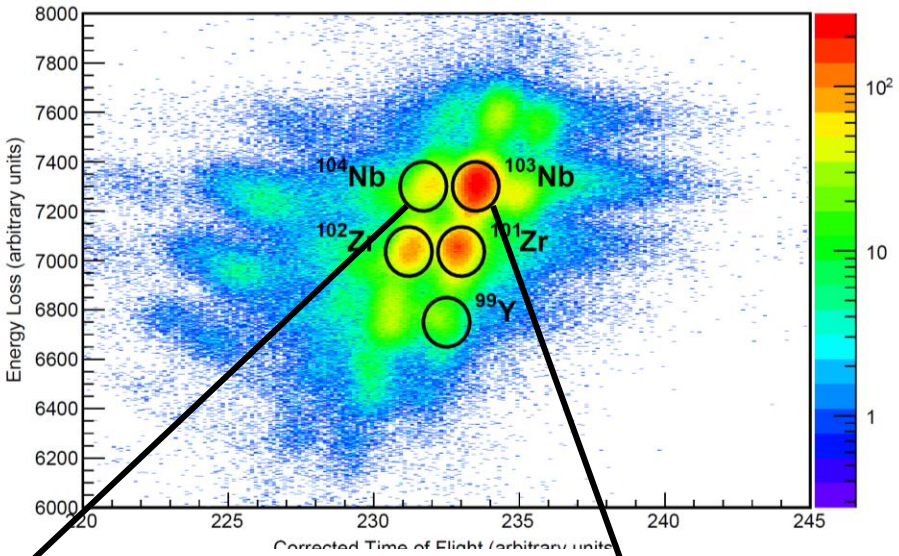


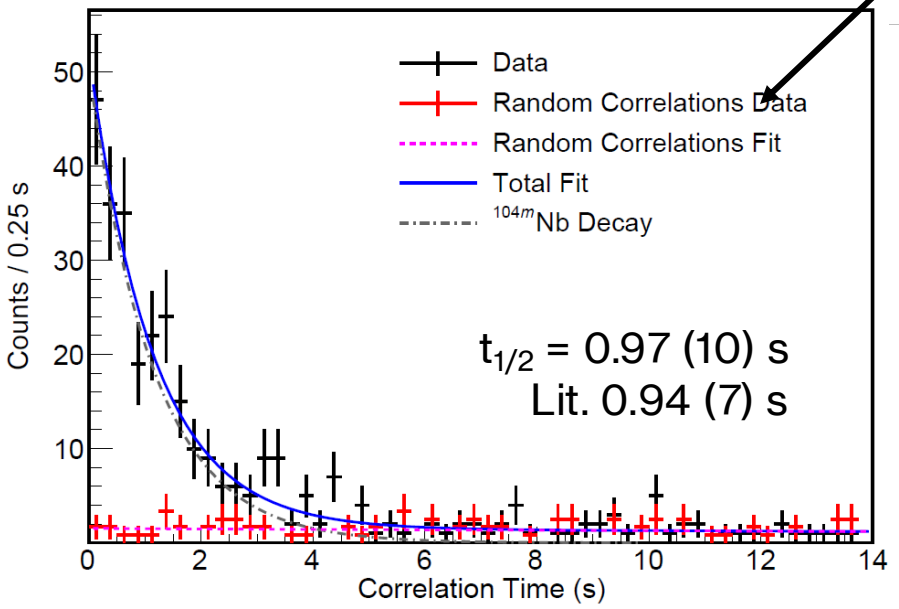


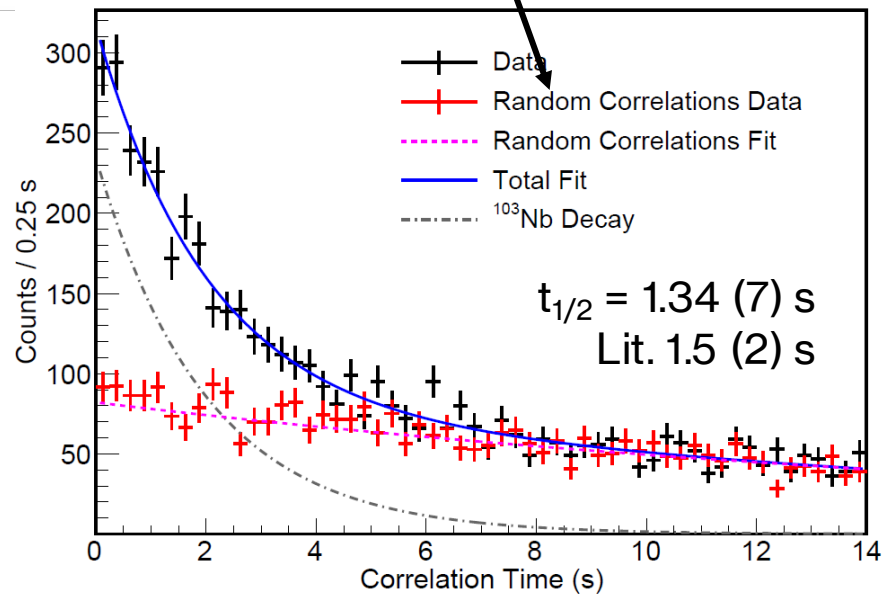
G. Lorusso, et. al. Phys. Rev. Lett. 114 (2015) 192501

How do you measure a half-life?







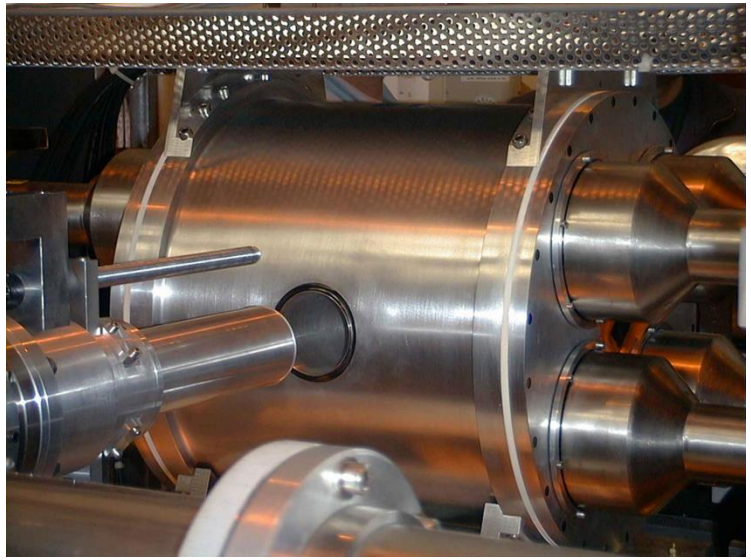


β Decay Intensity

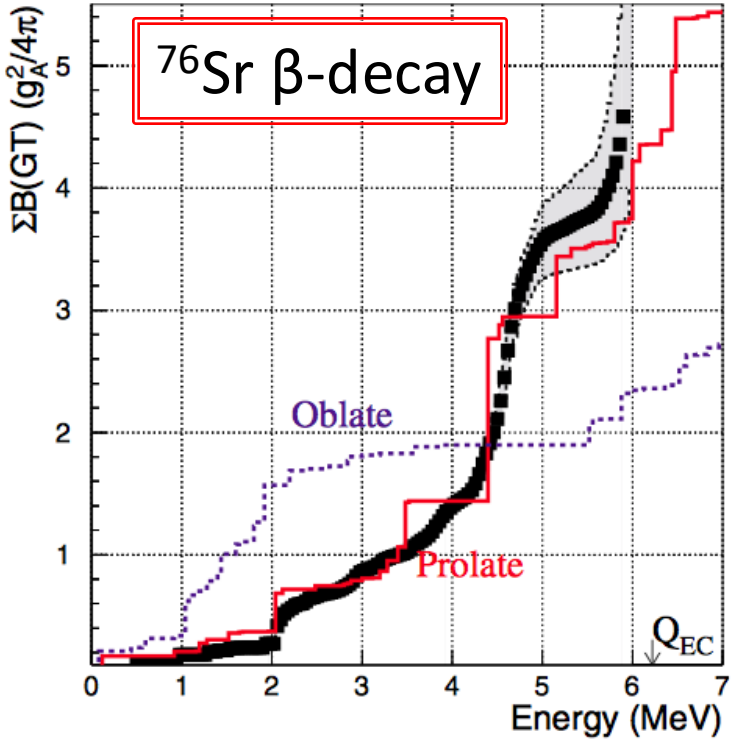
- Model constraints for better input in r-process calculations

  (Cannot measure everything - we need to rely on model predictions)

- Nuclear structure information
  - $T_{1/2}$ sensitive to nuclear shape
  - Can get same $T_{1/2}$ for different shapes
  - β-decay strength: sensitive constraint

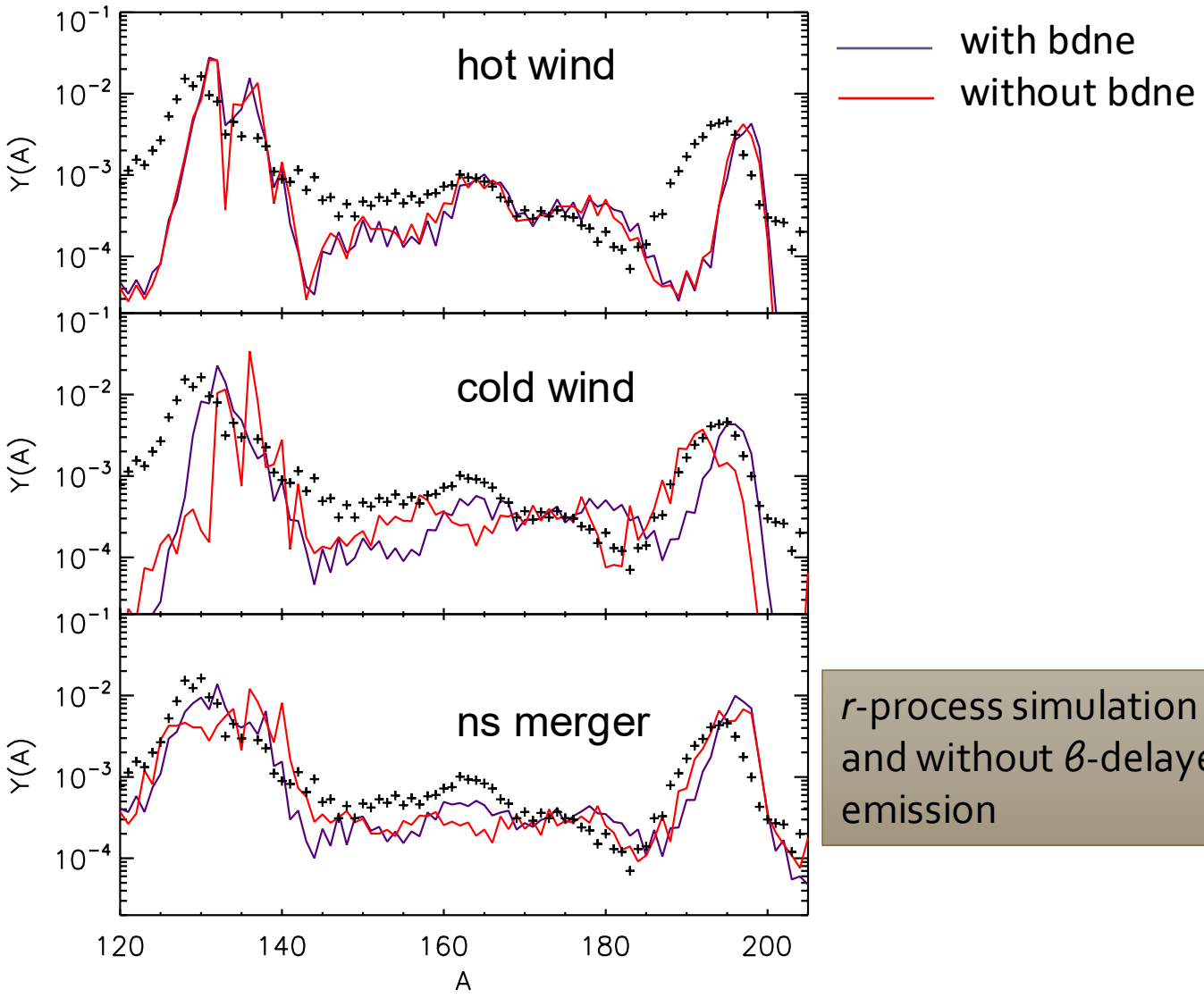


Lucrecia @ CERN



E. Nacher, et al., Phys. Rev. Lett. 92 (2004) 232501.

Importance of P_n Values

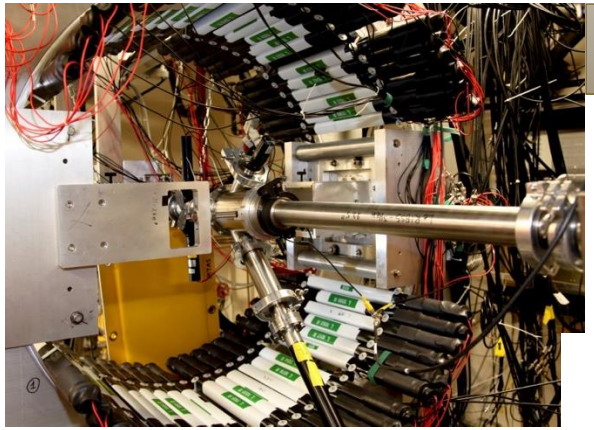


r-process simulation results with and without β-delayed neutron emission

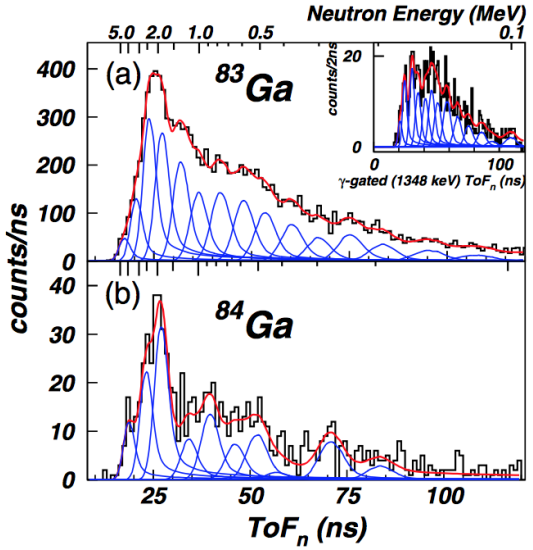
R. Surman, et. al. JPS Conf. Proc. 6 (2015) 010010

How do you measure them?

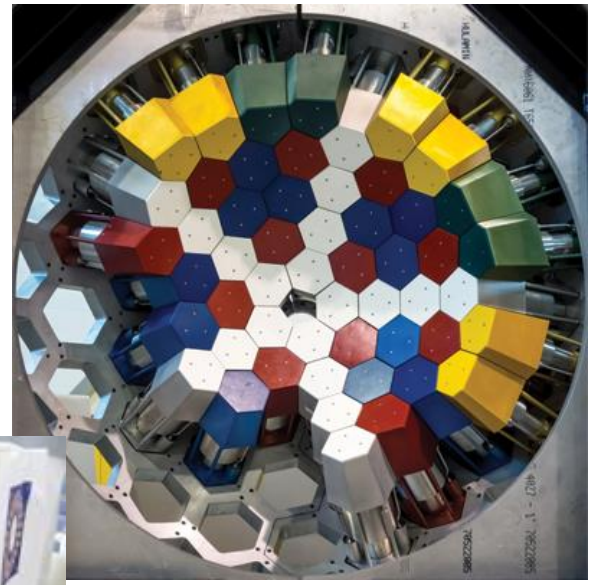
- Integrated measurement - ^{3}He counters (3Hen, NERO, BRIKEN)
- Time of flight – energy information (VANDLE, LENDA, DESCANT)



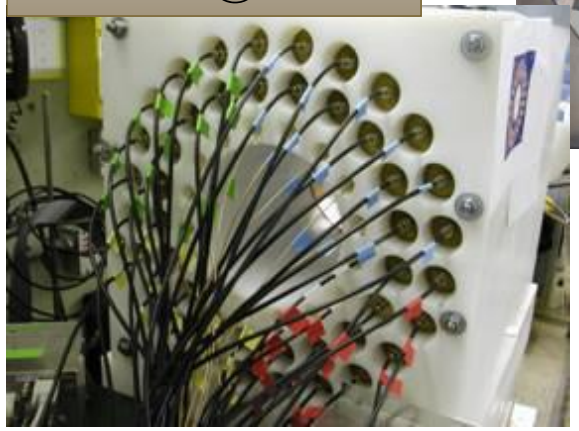
VANDLE @ ORNL



DESCANT @ TRIUMF

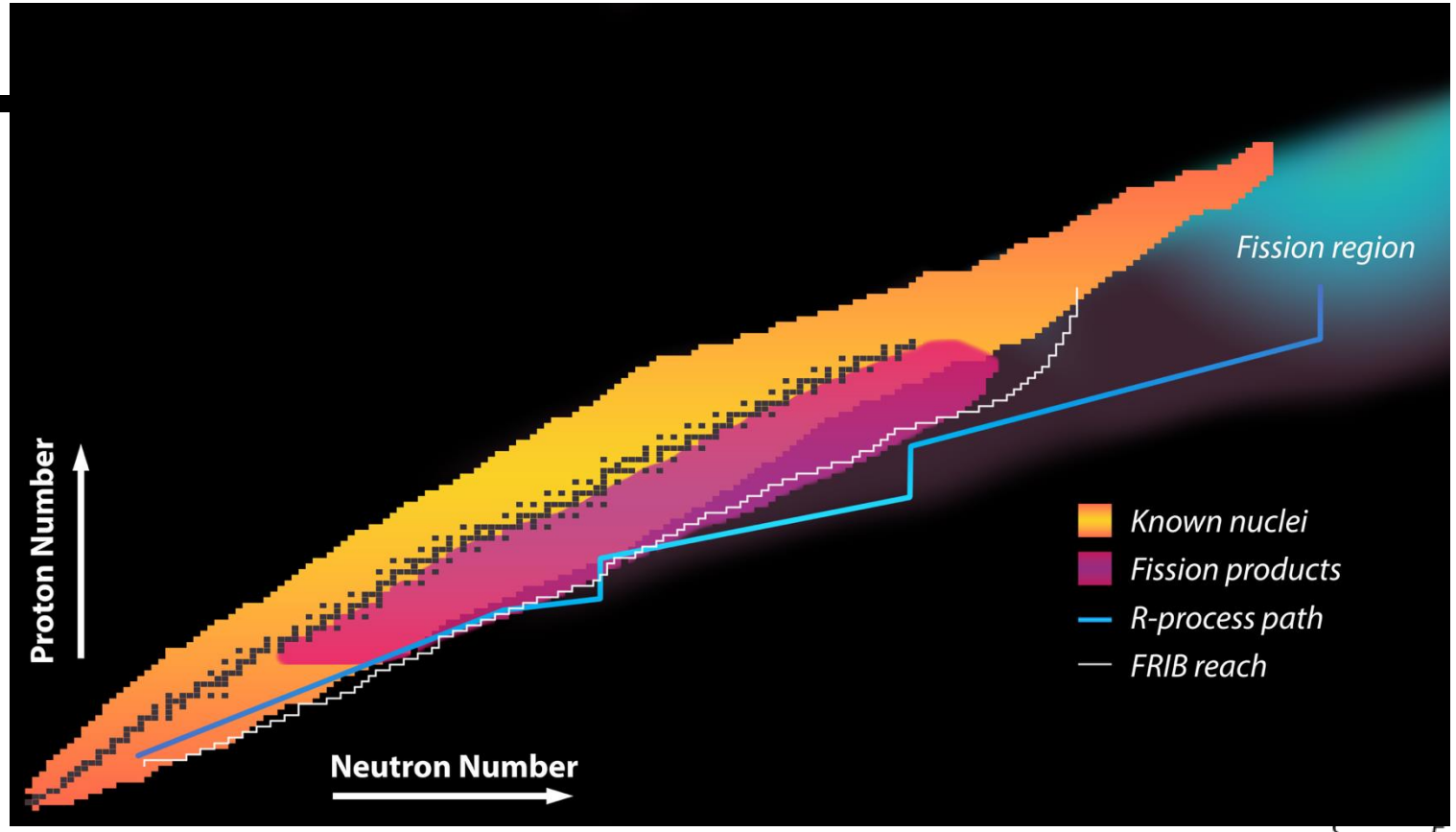


NERO @ MSU



M. Madurga, et. al. Phys. Rev. Lett. 117 (2016) 092502

Fission



Fission In R-process Elements

The FIRE collaboration explores the role of fission in the rapid neutron capture or r-process of nucleosynthesis

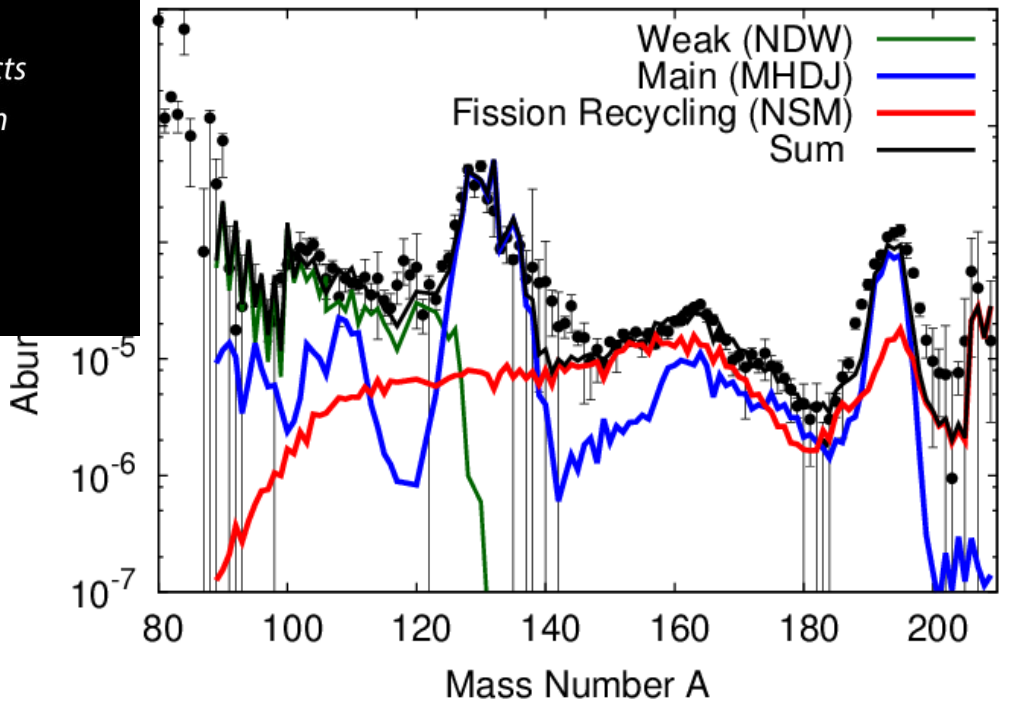
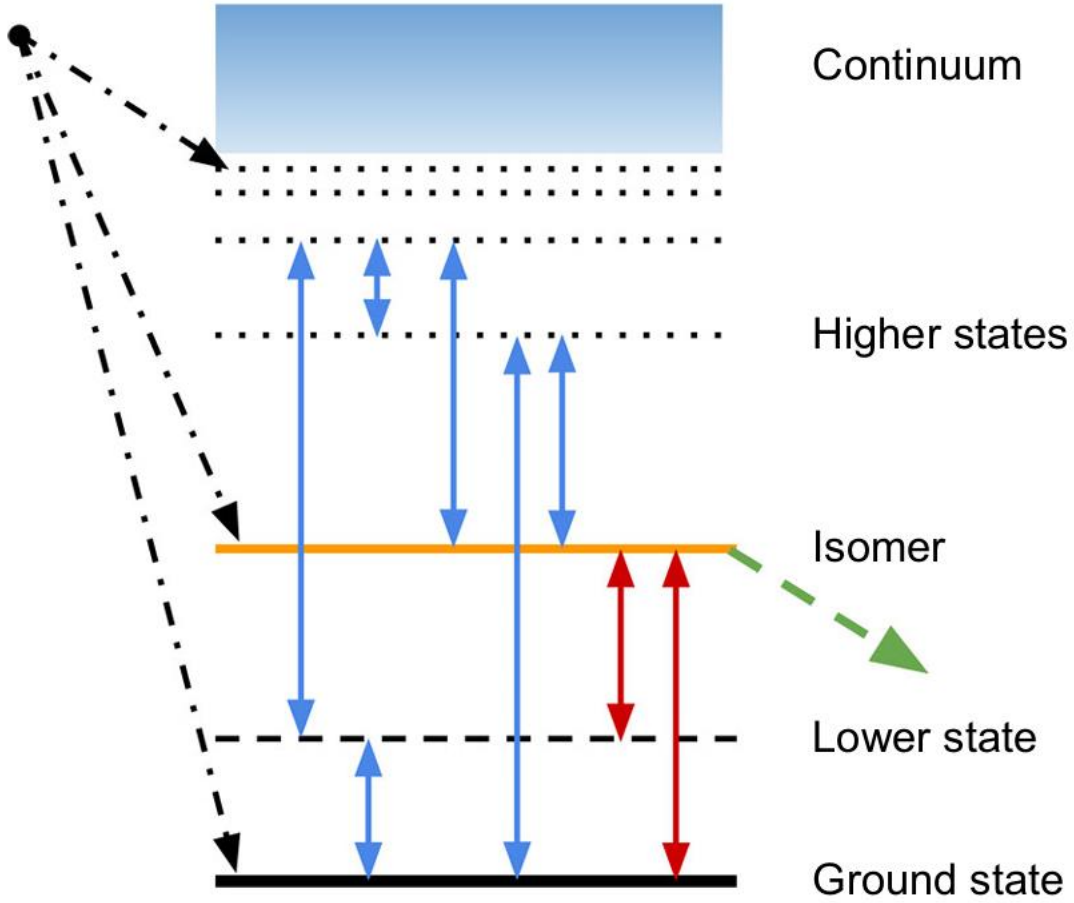
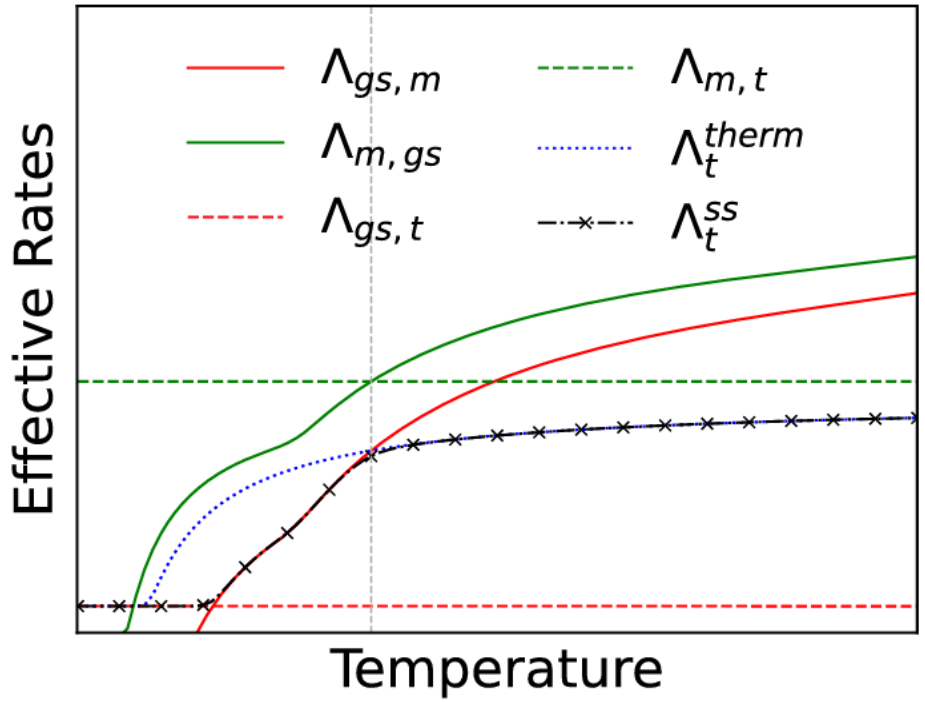


FIG. 3. (Color online) Average final abundance patterns for

Astromers

- Astrophysically-relevant isomers
- Very new studies!





G. Misch, Eur. Phys. J. Spec. Top. (2024) 233:1075–1099

Wrapping up!



Office Hours Friday 10AM, Edwards 204



Next class: Nuclear Astrophysics



HW#4 Due 11/13

Paper Outline of Final Due 11/18



Questions?

Final Exam Topics

- You should pick a topic from the following list (or propose something to me*). It cannot be closely related to your research. Topics must be approved by me.

- **Islands of Inversion**
- **Shape Coexistence -- Aashish**
- **Non-axial deformation**
- **The N=126 Magic Number**
- **Superheavy Nuclei – Noah (Z=126)**
- **Nuclear Fusion – Muhammad**
- **Reactor Neutrino Anomaly**
- **Neutrinoless double beta-decay -- Joel**
- **Beta-Delayed Particle Emission -- José**
- **s-process nucleosynthesis -- Amadie**
- **r-process nucleosynthesis – Ben**
- **rp-process nucleosynthesis**
- **Nuclear Pasta in Neutron Stars -- Nick**
- **Halo Nuclei -- Mark**
- **Nuclei and Medical Imaging -- Alan**
- **Radiation Therapy Techniques -- Nohora**
- **Medical Isotope Production -- Unish**
- **Effective Field Theory**
- **Rare Isotope Harvesting -- Cade**
- **Neutron Imaging -- Sushma**
- **Astromers**
- **Nuclear Masses and Astrophysics**
- **Nuclear Materials Analysis**
- **Density Functional Theories-**
- **Imaging Shapes of Nuclei -- Parker**
- **Alpha Clustering in Nuclei**
- **Nuclear Forensics* -- Eric**

Due Dates:
10/31 – Topic and at least 5 resources
11/18 – Outline of paper
11/25 – First Draft of Slides

Paper Outline

- Send me a PDF of your paper outline for the final on Nov. 18th by 5PM
- This should be a **detailed** outline
- NOT something like this:
  - I. Introduction
  - II. Experimental Methods
  - III. Analysis
  - IV. Results
  - V. Conclusion

Dissertation Outline (complete version by July 24)

1) Introduction
   a) Shell model and magic numbers
   b) 'island of inversion'
      i) Early experiments and theory – first indicator (masses), 2p-2h, rotational structures
      ii) Known properties of neutron-rich Mg isotopes
      iii) ^{33}Mg Literature Review
   c) Rotational Model / Collectivity – point to examples where this is successful in lighter nuclei
   d) Nilsson Model
   e) Multi-fragmentation?
   f) Beta Decay
      i) Selection Rules and log *ft*

2) Experimental Techniques
   a) Multi-fragmentation?
   b) Beta Decay
      i) Selection Rules and log *ft*
   c) Isotope Production and Identification
      i) NSCL Overview – CCF, A1900
      ii) NSCL experiment E11029
      iii) NSCL experiment E14063
   d) Experimental Vaults
      i) E11029 – S3 Vault
         (1) Two-stage fragmentation
         (2) S800 Spectrometer
            (a) Focal Plane detector suite
               (i) Calibrations
                  1. CRDCs
                  2. Ionization Chamber

Paper Guidelines

- 3 pages in two-column Physical Review C format
- **Paper Template**: https://www.overleaf.com/latex/templates/revtex-4-dot-2-template-and-sample/yydsrzvrqrzs
- You must use scientific/technical writing with correct grammar. You should make use of figures in the paper as well.
  - https://doi.org/10.1117/3.2317707.sup
  - https://cdn.journals.aps.org/files/rmpguapa.pdf
- Paper must include
  - Abstract
  - Introduction with clear motivation and background
  - Research Methods
  - Results
  - Discussion/Conclusion
  - References (at least 10)

This is meant to be a literature review, but I also want to see critical thinking.

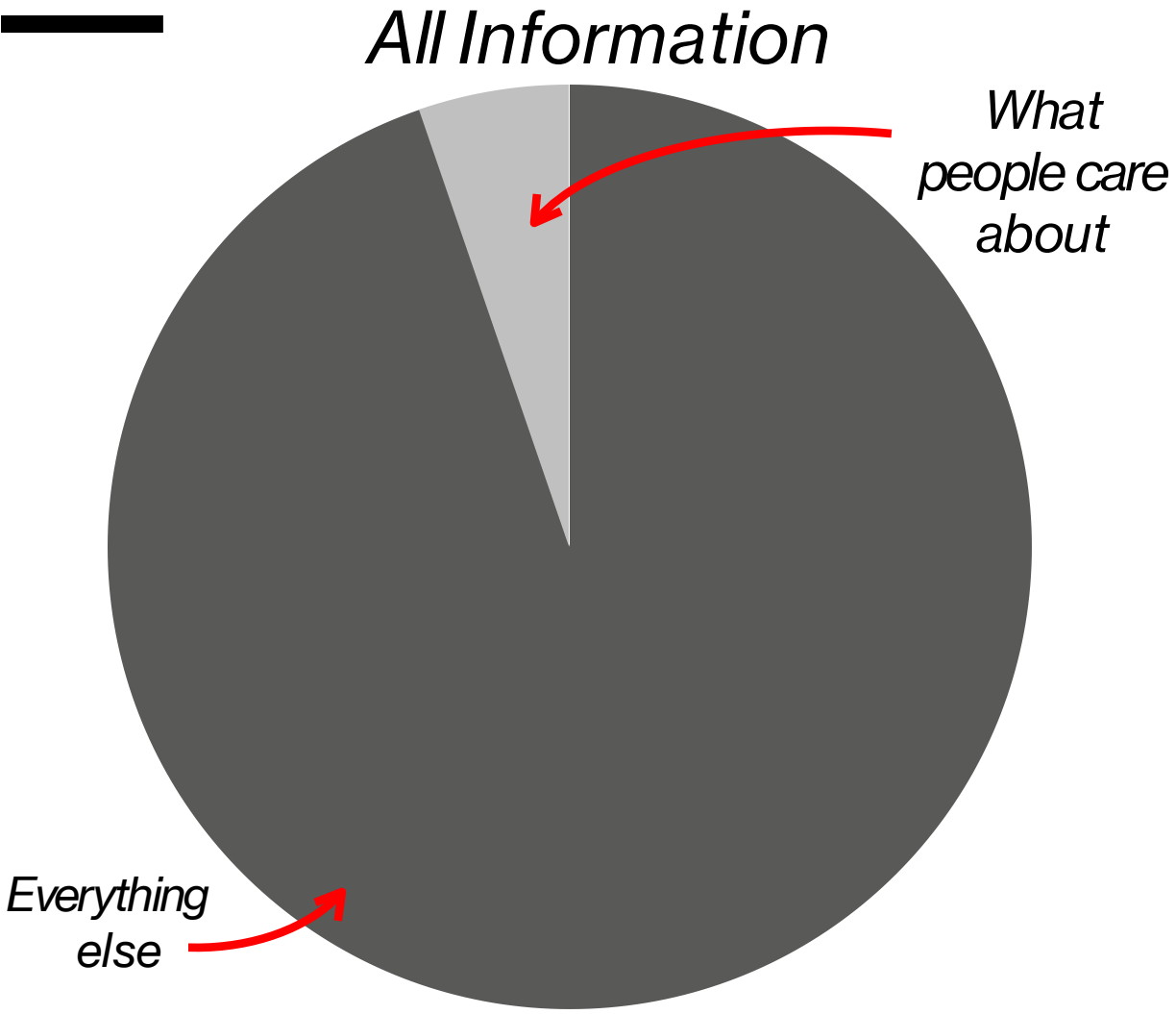
Look at recent PRC articles to get a sense of formatting and where to start

Paper is DUE at the start of the exam period on Dec. 9, 12:20PM



HOW TO GIVE A GOOD PRESENTATION

What is your story?



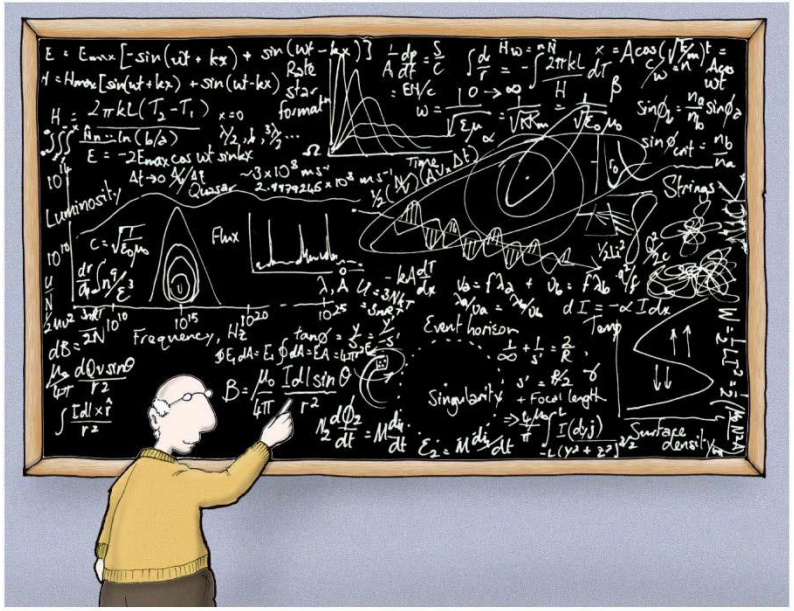
Presentations are all about the story.

Decide what story you want to tell and then what you need to be able to tell that story.

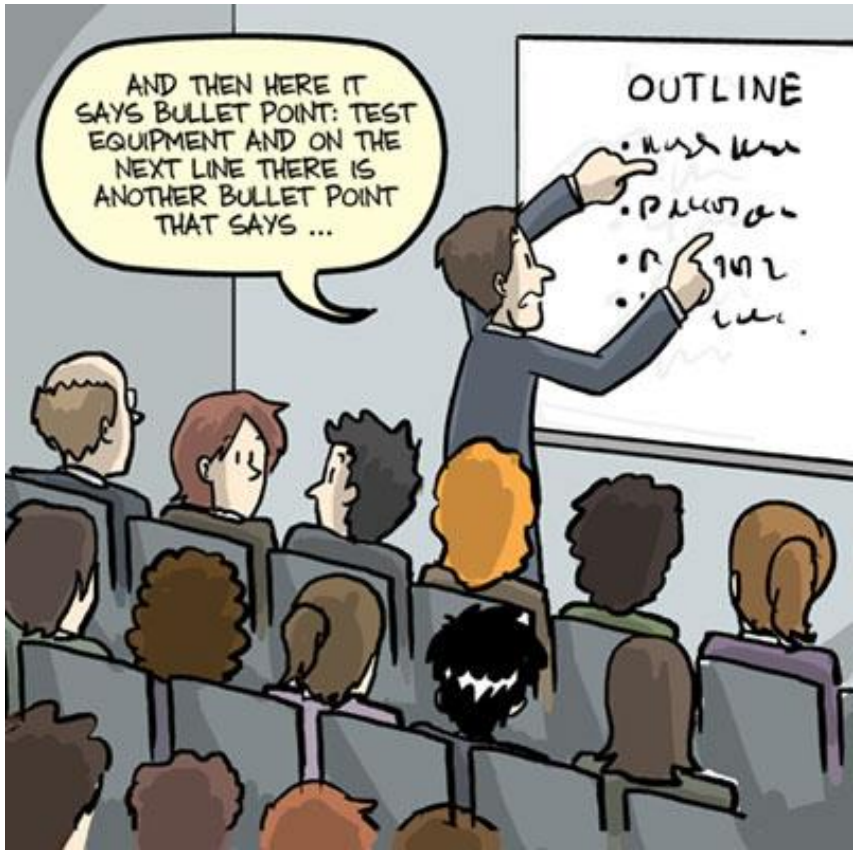
Deliver the highlights ... no one cares about the details (other than you and your advisor ☺)

Know Your Audience

- Tailor your presentation to the appropriate audience
- Who is your audience?
- Fellow graduate students (and me)

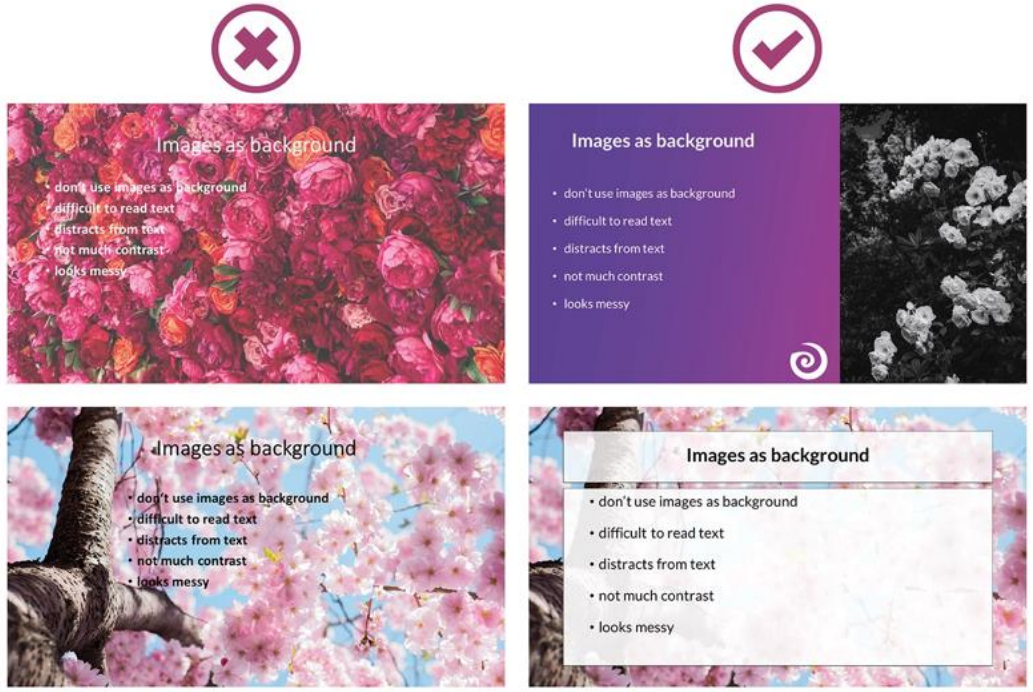


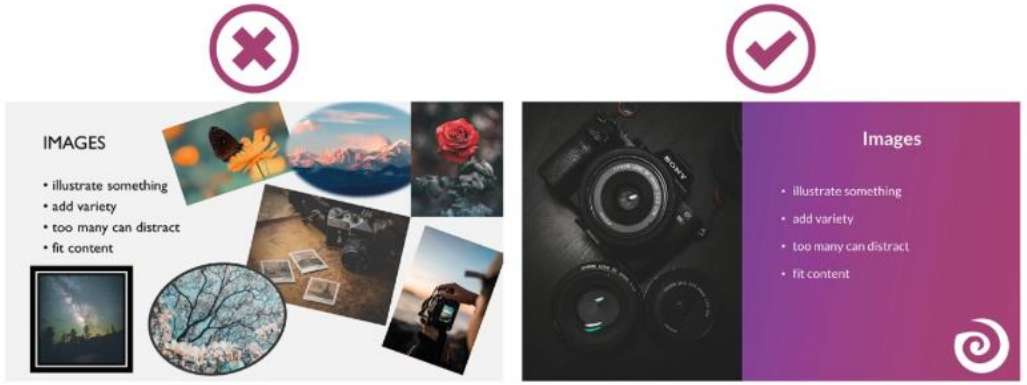
Astrophysics made simple



A Picture is Worth a Thousand Words

- Use figures to convey the story, but don’t let the use of figures overwhelm the message

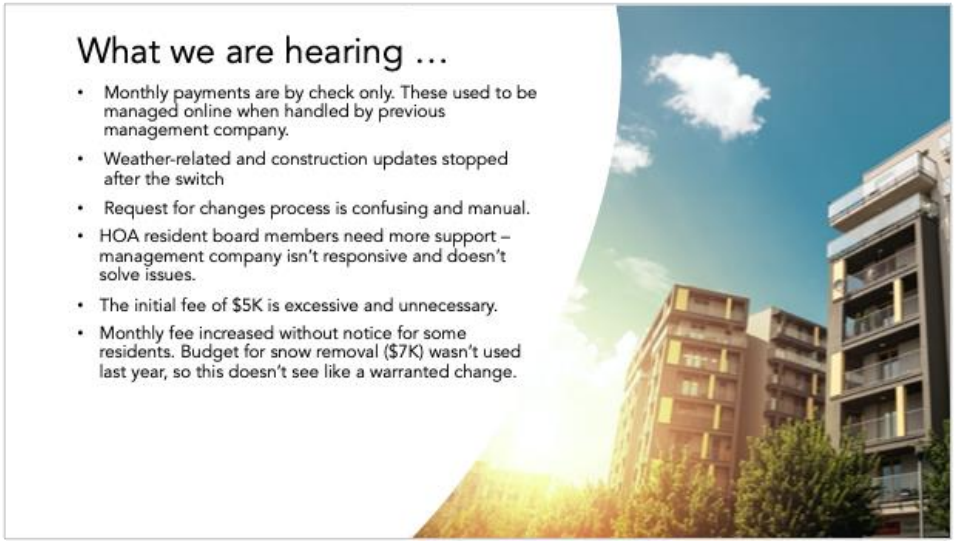




Avoid Excessively Wordy Slides

- This portion of the slide is worse than useless.
- It isn't possible for you to simultaneously read this and listen to me and to understand both.
- So I'm either wasting my time talking to you or wasting my time typing this.
- As a speaker I will likely fall into the trap of reading this text to you, simultaneously boring you and me, likely disengaging both of our brains.
- On top of all of that, this wall of text is completely hideous.
- The only **Acceptable Use of Bullets** contains:
    - Key facts/figures
    - Short, notable names/phrases

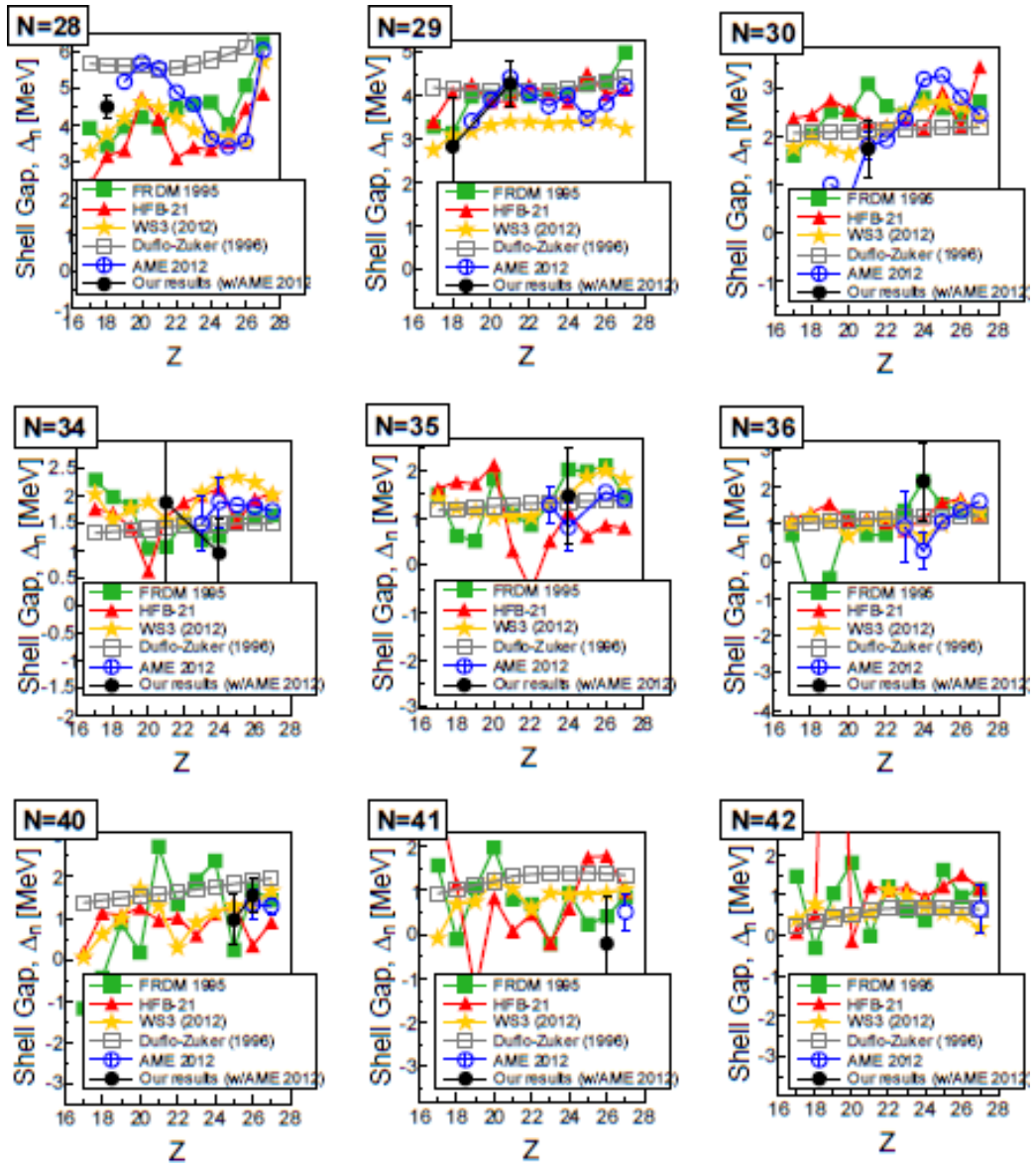
BEFORE: **a wordy slide**



AFTER: **storytelling with data**



Overly Small or Complex Figures Won't Add to the Story



Dense Tables Won't Add to the Story

Isotope	This experiment	AME2012	FRDM	HFB-21	DZ	WS3
^{48}Ar†	-22 280(310)	–	-21 240	-21 900	-21 889	-22 044
^{49}Ar†	-17 800(1100)	–	-14 880	-16 110	-15 648	-16 406
^{52}Sc	-40 300(520)	-40 167(142)	-39 360	-40 110	-39 241	-40 400
^{53}Sc	-38 170(570)	-38 107(270)	-36 840	-38 480	-37 148	-38 861
^{54}Sc	-33 750(630)	-33 599(363)	-32 030	-33 980	-32 623	-34 139
^{55}Sc	-30 520(580)	-29 977(463)	-29 170	-31 320	-29 597	-30 758
^{56}Sc†	$-24\ 850(590)^{+0}_{-540}$	–	-23 840	-25 230	-24 425	-25 146
^{57}Sc†	-21 000(1300)	–	-20 440	-22 550	-20 627	-21 115
^{59}Cr	-48 540(440)	-47 891(244)	-48 680	-49 160	-48 013	-48 451
^{60}Cr	-47 440(460)	-46 504(213)	-47 910	-48 200	-46 732	-46 779
^{61}Cr	-43 080(510)	-42 455(129)	-42 700	-43 710	-42 534	-42 461
^{62}Cr	-40 890(490)	-40 895(148)	-41 180	-41 960	-40 630	-40 445
^{63}Cr	-35 940(430)	-35 722(459)	-36 030	-37 290	-35 962	-35 773
^{64}Cr†	-33 480(440)	–	-34 950	-34 730	-33 545	-33 347
^{67}Mn†	-34 090(620)	–	-34 480	-34 960	-33 141	-33 294
^{67}Fe	-45 190(430)	-46 069(218)	-46 530	-46 940	-45 991	-45 577
^{68}Fe	-43 620(430)	-43 825(365)	-45 360	-45 170	-43 853	-43 665
^{69}Fe†	-39 350(600)	–	-40 230	-40 390	-39 156	-39 380

Excessive Equations Won't Add to the Story

Not OK

$$\varphi = \tan^{-1}\left[\frac{c\omega}{k - m\omega^2}\right] - \varphi_d$$

$$A = \frac{F_0 / m}{\sqrt{\left[\omega_0^2 - \omega^2\right]^2 + 4\gamma^2\omega^2}}$$

$$\gamma = \frac{c}{2m}$$

$$x(t) = A_h e^{-\gamma t}\sin(\omega' t + \varphi_h) + A\cos(\omega t - \varphi)$$

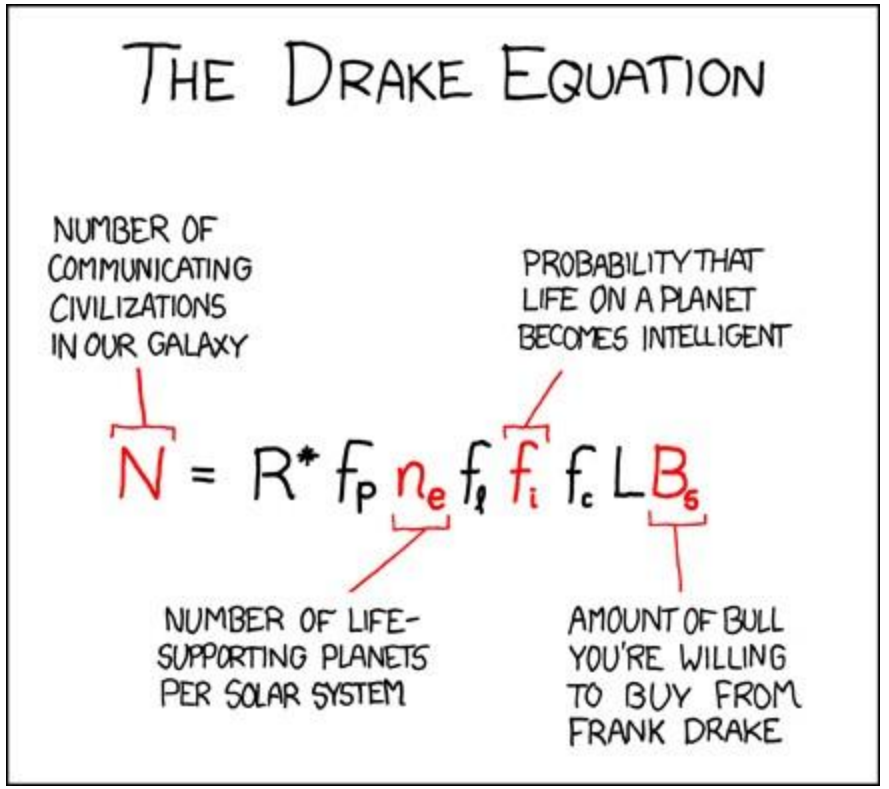
$$A_h = \frac{x_0 - A\cos\varphi}{\sin\varphi_h}$$

$$\omega' = \sqrt{\omega_0^2 - \gamma^2}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

$$\varphi_h = \tan^{-1}\left[\frac{\omega'(x_0 - A\cos\varphi)}{v_0 + \gamma(x_0 - A\cos\varphi) - A\omega\sin\varphi}\right]$$

OK

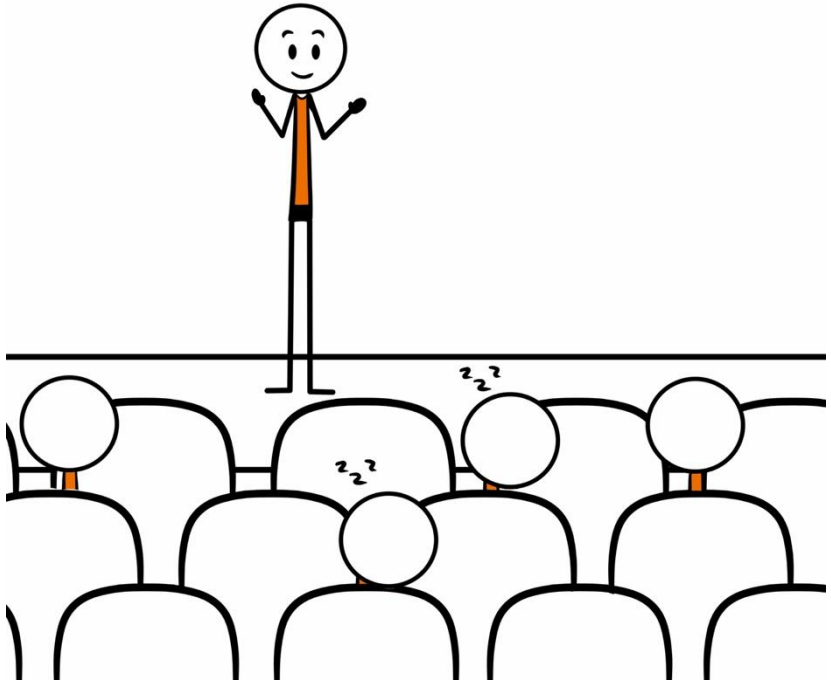


**only absolutely essential equations should be shown*

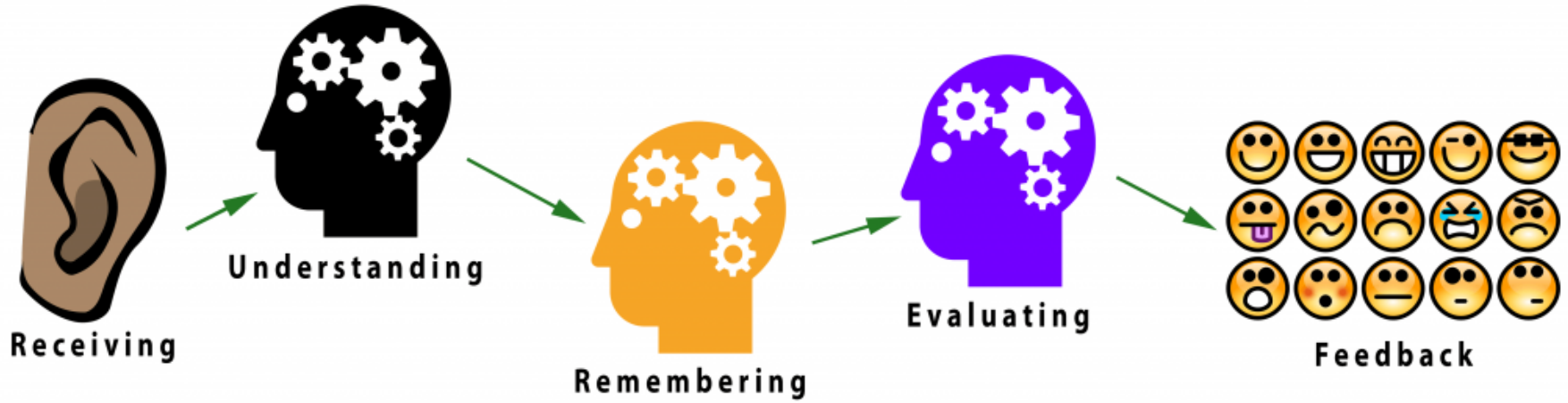
Finish On Time!

- You should always respect the allotted time
- You can achieve this by practicing your presentation ahead of time.
- Timing will be strictly adhered to in the final.





Active Listening



- 90% of communication is non-verbal
  - Eye contact
  - Nodding
  - Taking notes
- You should be doing this all the time to remain engaged in lectures, seminars, colloquia, etc.
- Peer reviews and questions are also part of the final!

General Things

- Make slides visually pleasing
- Check grammar, spelling, formatting
- Don’t make too many slides
  - Usually 1 slide/minute is the norm, but it depends on you.
- Cite use of all figures, tables, etc.
- Be careful with animations.
- Talk to your audience, not the slides
- Make eye contact with the audience
- Pace yourself
- Test your presentation ahead of the class

Presentation Guidelines

- Two groups (8 each):
  - Dec. 4: 2PM – **4PM**
  - Dec. 9: 12:20PM – 2:20PM
- Presentation should be ~12 slides and must include
  - Clear motivation
  - Overview of the topic
  - Discussion of Methods
  - Conclusion and Future Outlook
  - References

- **Presentation Length**: 10 min + 2 min for Q&A
- Timing will strictly be adhered to
- You should be prepared to answer audience questions with critical thinking
- Audience members must actively be engaged by asking questions and doing peer reviews. This is part of your grade!

Presentation Dates Volunteers

Dec. 4 2-4PM	Dec. 9 12:20 – 2:20PM
Unish	Joel
Mark	Muhammad
Noah	Amadie
Nohora	Sushma
Aashish	Nick
Cade	José
Eric	Ben
Alan	Parker